

VentureMiner AI

Monitoring & Observability

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Document 25 — Monitoring & Observability

How we see the platform. Logs, metrics, traces, RUM, synthetics, dashboards, alerts, and SLOs.

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1. Purpose & Scope

This document is the contract for observability. It defines what we collect, how we collect it, how we surface it, and how we act on it.

2. Principles

1. Three pillars by default. Every service emits logs, metrics, and traces.
2. Structured. JSON logs; standardized metric names; OTLP traces.
3. PII-safe. PII is redacted at the source.
4. SLO-driven. Alerts are SLO burn-rate based.
5. Cheap at the edges. Sampling where it doesn't hurt diagnosis.

3. Logs

- Format: JSON to stdout; shipped via Vector to Datadog.
- Required fields: timestamp, level, service, env, trace_id, span_id, workspace_id (where available), message, attrs.
- PII: redacted by the AGT-SAFETY filter; a redaction counter is emitted.

4. Metrics

- Standard: Prometheus client.
- Naming: .. (e.g. opportunity.created.count).
- Required service-level metrics:
 - requests_total (counter)
 - request_duration_seconds (histogram)
 - errors_total (counter, by code)
 - in_flight (gauge)
- AI plane adds:
 - llm.tokens.total (counter, by direction)
 - llm.cost_usd.total (counter)
 - mcp.calls.total (counter, by tool, by status)
 - verifier.pass_rate (gauge)

5. Traces

- Tool: OpenTelemetry → Datadog APM.
- Sampling: 100% for AI plane; 10% for HTTP at production; 100% in staging.
- Required attributes: workspace_id, user_id (hashed), run_id, agent_id (AI plane), tool_id (AI plane).
- Service map: built from spans; reviewed monthly.

6. Real User Monitoring

- Tool: Datadog RUM.
- Coverage: all web pages.
- PII: scrubbed; we do not capture form values.
- Sampling: 100% (cheap).

7. Synthetics

- API smoke: every 5 min; 10 critical endpoints.
- Browser smoke: every 30 min; key user journey.
- Multi-region: 3 PoPs.
- Failures page on-call.

8. Dashboards

Standard dashboards (one per service + cross-cutting):

- AI plane: per-agent latency, per-tool cost, verifier pass rate.
- Cost: per-service, per-workspace.

A new service must have a dashboard before it ships.

9. Alerts

- SLO burn rate — page on burn $\geq 2x$ in 1h; ticket on burn $\geq 1x$ in 24h.
- Error spike — page on error rate $> 5x$ baseline for 5 min.
- Saturation — page on resource $> 85\%$ for 15 min.
- Synthetics — page on any smoke fail.
- Cost — page on per-workspace cost $> 1.5x$ of trailing 7-day average.

Every alert has a runbook link and an owner team.

10. SLOs

Public API	success rate	99.9% (30d)
Dashboard	TTI $< 1.5s$ p75	99.5%
Report generation	< 10 min p95	99%
Discovery	$< 30s$ p95 for fresh hit	99%
Validation (Standard)	< 8 min p75	95%
AI verifier	pass rate	90%

SLOs are reviewed quarterly.

11. AI plane observability

See Document 18 §2.

12. Cost observability

- Per-workspace cost dashboards.
- Per-service cost dashboards.
- Cost anomaly detection (2σ over trailing 14 days).
- Reserved Instance / Savings Plan coverage tracked.

13. On-call

- Rotation: weekly; primary + secondary.
- Tooling: PagerDuty + Slack + Datadog on-call.
- Handoff: Friday 10:00 local; documented.
- Postmortem: within 7 days of any SEV-1 / SEV-2.

14.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

14.2 Cross-references

- TRD: Document 02 §11.
- DevOps: Document 23.
- AI Operations: Document 18.
- Operations Guide: Document 28.

End of Document 25 — Monitoring & Observability. If you can't see it, you can't fix it.